

Excluded-volume expansion of Archie's law for gas and solute diffusivities and electrical and thermal conductivities in variably saturated porous media

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[1] Describing and predicting gas and solute diffusivities and electrical and thermal conductivities under variably saturated fluid conditions are necessary for simulating gas, solute, and heat transport in soils. On the basis of comprehensive data for gas (D_p) and solute (D_s) diffusivities and electrical (EC) and thermal (TC) conductivities for differently textured and variably saturated soils, we investigated analogies and differences between the four parameters. At fluid (water or air) saturation, relative parameter values for D_p , D_s , and EC were all well described by an excluded-volume expansion of Archie's first law. The cementation exponent in Archie's first law was close to 1.5 for all parameters. At fluid-unsaturated conditions, relative values of D_p , D_s , and EC (normalized at fluid saturation) were well described by an excluded-volume expansion of Archie's second law. In the case of relative TC, the saturation exponent in Archie's second law was substituted by the inverse of it for the three other parameters since water bridge effects dramatically enhance the TC with increasing moisture contents in relatively dry porous media. If appropriate but different expressions for a percolation threshold in Archie's second law were applied for the four parameters, a saturation exponent value of around 2.0 generally gave accurate predictions of all four parameters for differently textured soils. Finally, the excluded-volume expansion of Archie's second law was modified to also represent porous media with bimodal pore size distribution and well-described data for D_p and D_s in aggregated soil.

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1. Introduction

[2] Knowledge of mass and energy transport phenomena in unsaturated soils is highly important in both the agricultural and engineering fields, and increasingly important in hydrology and climatology. Diffusive gas and solute transport in soils controls the fate and emissions of toxic gases and dissolved contaminants at polluted sites, the soil microbial activity through oxygen gas availability, and the plant uptake of nutrients in cultivated soils [Kruse *et al.*, 1996; Ball *et al.*, 1997; Jungk and Claassen, 1997; Poulsen *et al.*, 1998, 1999; van der Ploeg *et al.*, 1999]. The understanding of conductive heat transport is essential for many engineering applications such as thermal treatment of contaminated soil, ground freezing, underground energy

conservation, and buried electric cables [Ochsner *et al.*, 2001; Chen, 2008]. For example, an effective in situ thermal treatment of soils contaminated by volatile organic chemicals (e.g., tetrachloroethylene or gasoline vapors) requires the knowledge of simultaneous heat and gas transport characteristics in soils and thereby knowledge of gas diffusivity and electrical and thermal conductivities in variably saturated porous media [Ho *et al.*, 1995; Saito *et al.*, 1998; Merino and Bucalá, 2007]. Also, heat transport and thus thermal conductivity affect surface energy partitioning, and hence control plant growth factors such as seed germination, seed emergence, and establishments [Abu-Hamdeh and Reeder, 2000; Ekwue *et al.*, 2006].

[3] The diffusive gas and solute transport and conductive heat transport are governed by the gas diffusion coefficient (D_p , $\text{m}^2 \text{s}^{-1}$), the solute diffusion coefficient (D_s , $\text{m}^2 \text{s}^{-1}$), and the thermal conductivity (TC, $\text{W m}^{-1} \text{K}^{-1}$), respectively. The simultaneous knowledge of D_p and D_s in soils is useful in characterizing functional soil structure, hereunder characterizing liquid- and air-filled pore networks and their tortuosity and connectivity under variably saturated conditions [Moldrup *et al.*, 2001]. In addition to these three transport parameters, the soil electrical conductivity (EC, S m^{-1}) yields important information on not only soil salinity (i.e.,

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solute concentration) but also soil pore structure including pore network configuration and pore tortuosity in the soil fluid phases [Rhoades et al., 1976; Wildenschild et al., 2000].

[4] Previous studies have reported interrelations between EC, D_p , and TC measurements as well as a suggested direct relation between EC and D_s , and proposed predictive models for D_p , D_s , and TC based on EC models [Martys, 1999; Revil, 2000; Weerts et al., 2001; Boving and Grathwohl, 2001; Jounnot and Revil, 2010]. Since nondestructive, simple, and easy in situ EC measurements at large scale are feasible through geophysical survey methods such as electromagnetic induction (EMI) or direct current (DC) resistivity tomography, EC-based predictions of other transport parameters and their spatial distributions are highly promising [Lesch et al., 2005; Robinson et al., 2008]. Thus a comprehensive understanding of the relations between these four parameters (EC, D_p , D_s , and TC) is relevant for simulating mass (gas and solute) and energy (heat) transport through a multiphase (gas, liquid, and solid) porous media system, and for characterizing the connectivity and tortuosity of soil pore and particle networks. All four parameters are highly affected by pore structure and fluid content (φ , $\text{m}^3 \text{m}^{-3}$). Water content (φ_w , $\text{m}^3 \text{m}^{-3}$) is the controlling factor for EC and D_s , while air content (φ_{nw} , $\text{m}^3 \text{m}^{-3}$) is the controlling factor for D_p . The TC is affected by all soil phases [Lu et al., 2007] and thus requires a different approach.

[5] Archie [1942] presented the empirical relationships between EC and φ at variably saturated conditions, now known as Archie's first law at fluid saturation and Archie's second law at fluid-unsaturated condition [Martys, 1999; Revil and Jounnot, 2008]. The Archie's laws have been used to describe bulk (effective) diffusive and conductive transport parameters for porous media at variably saturated condition [Marshall, 1959; Millington, 1959; Mualem and Friedman, 1991; Martys, 1999; Moldrup et al., 2000; Boving and Grathwohl, 2001; Friedman, 2005]. At fluid saturation, a generalized Archie's first law for relative gas and solute diffusivities and electrical conductivity can be written as

$$\frac{P_{\text{sat}}}{P_0} = \varphi_{\text{sat}}^m, \quad (1)$$

where P_{sat} is the bulk parameter value at fluid saturation, P_0 is the parameter value in the pure fluid phase, φ_{sat} is the saturated fluid content (i.e., soil total porosity, $\text{m}^3 \text{m}^{-3}$), and m is the cementation exponent. For EC, Archie [1942] found m values around 1.3 for unconsolidated sands and 1.8–2.0 for consolidated sandstones. Brace et al. [1965] also reported the m of around 2.0 for EC data for sedimentary rocks. The m values for EC in various porous media were reviewed by Friedman [2005], with m typically ranging from 1.2 to 4.0. Friedman [2005] also showed smaller m values for glass spheres and sandy soils (1.3–1.8) as compared with sandstones and tuff (1.6–3.3). On the basis of theoretical assumptions representing given porous media networks, Millington [1959] and Marshall [1959] derived $m = 1.33$ and 1.5, respectively, for D_p . Boving and Grathwohl [2001] showed m ranging from 1.9 to 3.0 for D_s in limestones and sandstones.

[6] At fluid-unsaturated conditions, a generalized Archie's second law for gas and solute diffusivities and electrical conductivity can be written as

$$\frac{P}{P_{\text{sat}}} = \left(\frac{\varphi}{\varphi_{\text{sat}}} \right)^n, \quad (2a)$$

$$\frac{P}{P_0} = (\varphi_{\text{sat}})^m \left(\frac{\varphi}{\varphi_{\text{sat}}} \right)^n, \quad (2b)$$

where P is the bulk parameter value at fluid-unsaturated condition and n is the saturation exponent. Archie [1942] showed n values close to 2.0 for both consolidated and unconsolidated sands. Gorman and Kelly [1990] obtained average best fit values of $m = 1.30$ and $n = 1.53$ from electrical resistivity measurements for mixtures of Ottawa sand size fractions. Mualem and Friedman [1991] derived $m = 1.5$ and $n = 2.5$ for EC data based on the predictive model for hydraulic conductivity [Mualem, 1976]. Moldrup et al. [2000] proposed the water-induced linear reduction (WLR) model for D_p , based on and extending the classic models for dry soils by Penman [1940], Marshall [1959], and Millington [1959]. In the WLR (Marshall) model, m and n can also be expressed as 1.5 and 2.5, respectively. Several previous studies have implied the applicability of the Archie's first and second law expressions for predicting either EC, D_p , or D_s at variably saturated condition, while few studies have considered the applicability of the Archie's law for TC data. Further, the possible correlations and analogies among all four parameters have not been investigated simultaneously, hereunder looking for possible similarities in the cementation exponent (m) and saturation exponent (n).

[7] In this study, on the basis of comprehensive data sets for gas and solute diffusion coefficients and electrical and thermal conductivities for differently textured and variably saturated soils from literature and own measurements, the analogies between the four parameters were investigated. Modified Archie's first and second laws considering percolation thresholds were developed and tested against the data sets for all four parameters including TC. Finally, the Archie's law-based models were also extended to represent bimodal pore structure and tested against D_p and D_s data for differently textured soils including aggregated soils.

2. Model Development

[8] In this study, the D_p , D_s , [EC-EC_s], and [TC-TC_s] were considered as the bulk soil parameter values (P) for each of the four considered transport parameters in soils. The EC_s (S m^{-1}) is the electrical conductivity at very dry conditions (almost zero water content), representing the EC through the water layer coating the surface of the grains (i.e., Stern layer and diffuse layer surface conductivities) [Bolève et al., 2007; Leroy et al., 2008], and TC_s ($\text{W m}^{-1} \text{K}^{-1}$) is the thermal conductivity through a solid phase, respectively (Table 1). For gas and solute diffusion coefficients and electrical conductivity in the pure fluid phase (P_0), we used gas diffusion coefficient in air ($D_{0,g}$), solute diffusion coefficient in water ($D_{0,l}$), and electrical conductivity in water (EC_w) (Table 1).

[9] Equations (1), (2a), and (2b) for each parameter do not consider the percolation threshold at which diffusive and

Table 1. Model Parameters Considered in This Study

	φ ($\text{m}^3 \text{m}^{-3}$)	φ_{th}^* ($\text{m}^3 \text{m}^{-3}$)	P/P_0	P/P_{sat}	φ_{th} ($\text{m}^3 \text{m}^{-3}$)
Electrical conductivity (EC)	φ_{W}	$= \varphi_{\text{th}}$	$(\text{EC}-\text{EC}_s)/\text{EC}_w$	$(\text{EC}-\text{EC}_s)/(\text{EC}_{\text{sat}}-\text{EC}_s)$	$0.8\rho_b\{(CF/2.7)+(OM)/1.0\}$
Gas diffusion coefficient (D_p)	φ_{nW}	0	$D_p/D_{0,g}$	$D_p/D_{p,\text{sat}}$	$0.1\varphi_{\text{sat}}$
Solute diffusion coefficient (D_s)	φ_{W}	$= \varphi_{\text{th}}$	$D_s/D_{0,1}$	$D_s/D_{s,\text{sat}}$	$0.8\rho_b\{(CF/2.7)+(OM)/1.0\}$
Thermal conductivity (TC)	φ_{W}	NG^a	NG^a	$(\text{TC}-\text{TC}_s)/(\text{TC}_{\text{sat}}-\text{TC}_s)$	$0.4\rho_b\{(CF/2.7)+(OM)/1.0\}$

^aNot given.

conductive transport diminish due to disconnected fluid phase. The existence of a percolation threshold for D_p and D_s at variably saturated condition has been reported in many studies [Freijer, 1994; Olesen *et al.*, 1999; Moldrup *et al.*, 2007; Resurreccion *et al.*, 2007, 2008a; Revil and Jounot, 2008]. On the basis of the numerical simulations for ion diffusion in porous media using the lattice Boltzmann method, Martys [1999] showed that Archie's second law (equations (2a) and (2b)) does not describe well ion diffusivity at lower water saturation approaching a percolation threshold.

[10] A modified Archie's first law considering a percolation threshold has been suggested by Sen *et al.* [1981], Balberg [1986], and Hunt [2004a] as

$$\frac{P_{\text{sat}}}{P_0} = (\phi_{\text{sat}} - \phi_{\text{th}}^*)^m, \quad (3)$$

where φ_{th}^* represents the inactive fluid phase for the given transport process at fluid saturation ($\text{m}^3 \text{m}^{-3}$).

[11] Considering a percolation threshold for each parameter at fluid-unsaturated conditions, equations (2a) and (2b) can be modified to form an excluded-volume expansion of Archie's second law,

$$\frac{P}{P_{\text{sat}}} = \left(\frac{\phi - \phi_{\text{th}}}{\phi_{\text{sat}} - \phi_{\text{th}}} \right)^n, \quad (4a)$$

$$\frac{P}{P_0} = (\phi_{\text{sat}} - \phi_{\text{th}}^*)^m \left(\frac{\phi - \phi_{\text{th}}}{\phi_{\text{sat}} - \phi_{\text{th}}} \right)^n, \quad (4b)$$

where φ_{th} is the percolation threshold at fluid-unsaturated condition ($\text{m}^3 \text{m}^{-3}$).

2.1. Application of Archie's Second Law to Thermal Conductivity ("Mirror Image" Approach)

[12] Figure 1 shows the TC and D_p values normalized by each parameter value at fluid saturation for sands as a function of φ normalized by φ_{sat} (i.e., $\varphi/\varphi_{\text{sat}}$). The TC and D_p data for sands with different size fractions are from Chen [2008] and Hamamoto *et al.* [2009a], respectively. The soil air content (φ_{nW}) was used as φ for D_p , while the volumetric water content (φ_{W}) was used for TC. The normalized D_p increased rapidly at higher φ due to the increase of pore continuity at dry conditions. Conversely, the normalized TC increased rapidly at lower φ (i.e., dry conditions) and the incremental increase in normalized TC decreased at higher φ . At dry condition, water bridges begin to form between soil particles, and the TC starts to increase rapidly because of the improved thermal contact between particles [Lu *et al.*, 2007; Ewing and Horton, 2007]. At wetter condition (i.e., higher φ), the incremental increase in TC becomes smaller because the water film becomes thicker and the increase in

TC with increasing soil water content depends largely on the displacement of air by water. The water bridge effects (i.e., enhanced thermal conduction through a solid phase) on TC were well described by considering the saturation exponent (n) in equation (4a) as the inverse of the n for the three other parameters (Figure 1). Revil [2000] has also discussed a mirror image relation between TC and EC using a differential effective medium scheme. A modified Archie's second law considering a percolation threshold for TC can thereby be written as

$$\frac{P}{P_{\text{sat}}} = \left(\frac{\phi - \phi_{\text{th}}}{\phi_{\text{sat}} - \phi_{\text{th}}} \right)^{1/n} \quad (5a)$$

$$\frac{P}{P_0} = (\phi_{\text{sat}} - \phi_{\text{th}}^*)^m \left(\frac{\phi - \phi_{\text{th}}}{\phi_{\text{sat}} - \phi_{\text{th}}} \right)^{1/n}. \quad (5b)$$

In comparison with for example D_p (Figure 1), the n in equations (4a) and (4b) represents water blockage effects on D_p , while the n in equations (5a) and (5b) represents water bridge effects on TC. For TC, higher n values indicate larger water bridge effects, resulting in a rapid increase of TC with increasing fluid (water) content at dry conditions. As shown in Figure 1 (assuming zero percolation threshold, $\varphi_{\text{th}} = 0$), this yields a "mirror image" curve for TC (equations (5a)

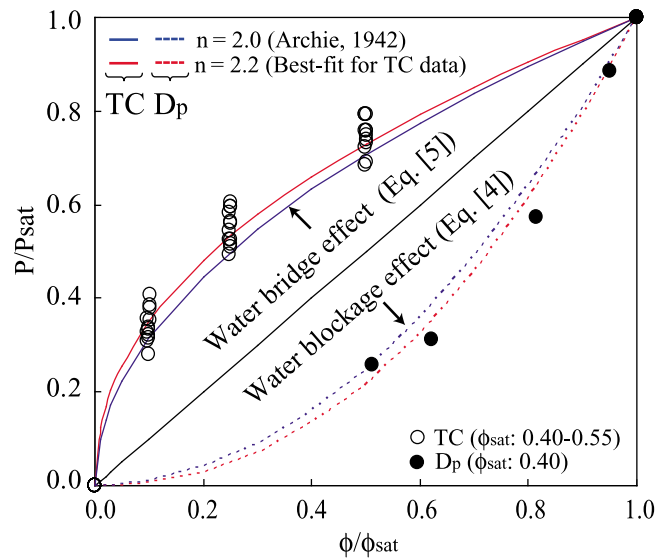


Figure 1. Normalized thermal conductivity (TC) and D_p values as a function of normalized φ . The solid and dotted lines represent the predictive curves for TC by equation (5a) and D_p by equation (4a), using $n = 2.0$ [Archie, 1942] and $n = 2.2$ (best fit for TC data).

Table 2. Data Sets for EC, D_p , D_s , and TC Used in This Study

		Number of Data Sets	Bulk Density (g cm ⁻³)	Clay Content (g g ⁻¹)	Reference
EC	Sand	6	1.37–1.84	0.00–0.10	<i>Rhoades et al.</i> [1976], <i>Weerts et al.</i> [1999, 2001], <i>Nadler et al.</i> [1984]
	Soil	7	1.32–1.61	0.06–0.17	
D_p	Sand	6	1.40–1.60	0.00–0.03	<i>Moldrup et al.</i> [2000], <i>Hamamoto et al.</i> [2009a]
	Soil	10	1.05–1.55	0.07–0.38	
D_s	Sand	5	1.48–1.58	0.00–0.03	<i>Olesen et al.</i> [1999], <i>Hamamoto et al.</i> [2009b]
	Soil	7	1.05–1.76	0.11–0.38	
TC	Sand	9	1.58–1.60	0.01–0.05	<i>Lu et al.</i> [2007], <i>Chen</i> [2008], <i>Jury and Horton</i> [2004]
	Soil	10	1.20–1.39	0.09–0.32	

and (5b)) as compared with the three other parameters including D_p (equations (4a) and (4b)) (discussed later).

2.2. Percolation Threshold

[13] The φ_{th} value for D_p was chosen according to *Hunt* [2004b] who proposed $0.1\varphi_{sat}$ based on fractal analysis for differently textured soils. In this study, we assumed that all air-filled pore spaces are well connected at air saturation (i.e., $\varphi_{th}^* = 0$ for D_p). Assuming a close relationship between the soil surface area and the film water thickness around soil/clay particles at relatively dry conditions, the volumetric content of fines (clay plus organic matter) was used to estimate φ_{th} for [EC-EC_s] and D_s following *Moldrup et al.* [2007]. They found $\varphi_{th} = 0.8\rho_b\{(CF/2.7)+(OM/1.0)\}$ (where ρ_b is a bulk density (Mg m⁻³), CF is a clay mass fraction, and OM is an organic matter content) to accurately represent the percolation threshold for D_s for seven variably textured and variably compacted soils. We further assumed the percolation thresholds for [EC-EC_s] and D_s maintain an inactive water content across moisture conditions including at water saturation (i.e., $\varphi_{th}^* = \varphi_{th}$) since even at water saturation low electromigration mobility in the Stern layer around clay particles may cause inactive water-filled pore spaces for EC and D_s [*Jougnot et al.*, 2009].

[14] We also assume the existence of a percolation threshold (φ_{th}) for [TC-TC_s] due to entrapped water (residual water) at dry condition, the degree of which is expected to be related with soil surface area (i.e., fines content). As a preliminary assumption, a reduced volumetric content of fines was assumed as a φ_{th} for [TC-TC_s]. When half of φ_{th} for D_s and [EC-EC_s] (i.e., $\varphi_{th} = 0.4\rho_b\{(CF/2.7) + (OM/1.0)\}$) was applied, the increase of normalized TC, $(TC-TC_s)/(TC_{sat}-TC_s)$, with increasing normalized φ , $(\varphi-\varphi_{th})/(\varphi_{sat}-\varphi_{th})$, was accurately described (also see Figure 4d). The reduced φ_{th} value for [TC-TC_s] as compared with [EC-EC_s] and D_s appears reasonable because thermal conductivity is likely not affected by the electrochemical characteristics of clay surfaces and soil water, and also small amounts of water may form the water bridges between soil particles at dry conditions, causing the enhancement of TC. The suggested percolation thresholds for all four parameters are summarized in Table 1.

2.3. Application of Archie's Second Law to Bimodal Porous Media

[15] Aggregated soils with a bimodal pore structure, which are generally formed by complex interrelationships of physical, biological, and chemical reactions such as a clay flocculation or cementation due to root exudations, also exhibit bimodal characteristics of transport parameters.

This is well known for D_p as a function of soil air content (φ_{nw}) [*Currie*, 1961; *Resurreccion et al.*, 2008b, 2010], and *Resurreccion et al.* [2010] proposed a two-independent-region (TIR) $D_p/D_{0,g}$ model for aggregated soils by applying cumulative power law functions ($H\varphi_{nw}^X$) to the inter-aggregate and intra-aggregate pore regions. They found that the coefficient H for the interaggregate region was best described by a nonlinear, water-content-dependent expression taking into account water blockage effects, while constant H (around 0.5) and X (=1) values typically sufficed for the intra-aggregate region. Similar to this approach, the modified second Archie's second law (equations (4a) and (4b)) was extended to represent bimodal transport characteristics,

$$\frac{P}{P_0} = (\phi_1 - \phi_{th}^*)^m \left(\frac{\phi - \phi_{th}}{\phi_1 - \phi_{th}} \right)^n, \text{ for } \varphi \leq \varphi_1 \quad (6a)$$

$$\frac{P}{P_0} = (\phi_1 - \phi_{th}^*)^m + H(\phi - \phi_1), \text{ for } \varphi_1 < \varphi \quad (6b)$$

where φ_1 is the interaggregate porosity (m³ m⁻³). Again, we would assume the use of $1/n$ instead of n in equation (6a) for TC, but this was not tested due to lack of data.

2.4. Statistical Analysis

[16] The root-mean-square error (RMSE) was used to evaluate model performances of the modified Archie's second law (equations (4a), (4b), (5a), and (5b)) against EC, D_p , D_s , and TC data. The RMSE was calculated from

$$RMSE = \sqrt{\frac{1}{s} \sum_{i=1}^s (d_i)^2}, \quad (7)$$

where d_i is the difference between the predicted and the measured values of the parameters at a given fluid content and s is the number of measurements.

3. Material and Methods

3.1. Soils and Data

[17] In the analysis, we used data sets for EC, D_p , D_s , and TC for variably saturated and variably textured soils from the literature as well as our own D_p measurements for sand (Hokkaido sand) and loamy soil (Hokkaido loam). The number of data sets for each parameter and ranges of bulk density and clay content for sands and loamy-clayey soils (labeled as "soil" in this paper) and main references for each parameter data set are presented in Table 2.

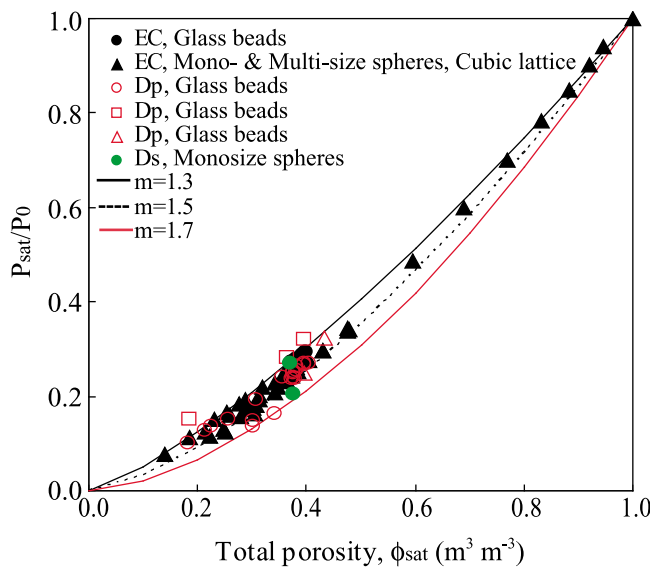


Figure 2. Relative parameter value, P/P_0 , for EC, D_p , and D_s for artificial materials at fluid saturation. The solid and dotted lines represent the predictive curves by equation (1). Data are from Friedman [2005], solid black circle; Robinson and Friedman [2005], solid black triangle; Currie [1960b], open red circle; Penman [1940], open red square; Papendick and Runkles [1965], open red triangle; and Robinson and Friedman [2005], solid green circle.

[18] The following data sets were used for the EC, D_p , and D_s data at fluid saturation. EC data: two artificial porous media from Friedman [2005] and Robinson and Friedman [2005], and one sand and two soils from Mualem and Friedman [1991], Rhoades et al. [1976], and Weerts et al. [1999]; D_p data: three artificial materials (glass beads) from Currie [1960b], Penman [1940], and Papendick and Runkles [1965], and four sands from Hamamoto et al. [2009a], Currie [1960b], Penman [1940], and Taylor [1949], and water-filled aggregate (calcined clay) from Jones et al. [2003]; D_s data: one artificial material from Robinson and Friedman [2005], and one sand from Hamamoto et al. [2009b].

[19] The EC, D_p , D_s , and TC data for sands and soils at fluid-unsaturated conditions were obtained from literature as follows. EC data: six sands from Nadler et al. [1984], Persson [1997], Weerts et al. [1999, 2001], Amente et al. [2000], and Tuli and Hopmans [2004], and seven soils from Rhoades et al. [1976], Nadler [1982], Nadler et al. [1984], Mallants et al. [1996], Persson and Berndtsson [1998], and Tuli and Hopmans [2004]; D_p data: six sands from Currie [1961], Osozawa [1998], Moldrup et al. [2000], and Hamamoto et al. [2009a], and 10 soils from Shearer et al. [1973], Petersen et al. [1994], Jin and Jury [1996], Moldrup et al. [2000, and references therein], and Moldrup et al. [2005, and references therein]; D_s data: five sands from Kaselowsky [1990], Mehta et al. [1995], and Hamamoto et al. [2009b], and six soils from Rowell et al. [1967], Kaselowsky [1990], Bhadoria et al. [1991], Olesen et al. [1999], and Hamamoto et al. [2009b]; TC data: nine sands from Jury and Horton [2004, and references therein], Lu et al. [2007], and Chen [2008], and 10 soils from Jury

and Horton [2004, and reference therein], and Chen [2008]. The D_p and TC data for organic soils were also considered in this study: D_p and TC data from Lauren et al. [2000], and TC data from Sharratt [1997], Bonan [2002], and Jury and Horton [2004, and references therein].

3.2. Measurements of Soil Gas Diffusion Coefficient

[20] The D_p was measured for repacked Hokkaido loam and Hokkaido sand. Hokkaido loam (Gray lowland soil) and Hokkaido sand were sampled from the surface layer in agricultural field at the campus of Hokkaido University and at Ishikari-city in Hokkaido, Japan, respectively. All soils were sieved (2 mm mesh) and repacked into 100 cm³ core samples (5.6 cm diameter and 4.06 cm height) with a given bulk density ($\rho_b = 1.05$ g cm⁻³ for Hokkaido loam, $\rho_b = 1.44$ g cm⁻³ for Hokkaido sand) according to Hamamoto et al. [2009b]. For Hokkaido sand, the soil air content in the core was adjusted by repacking each soil sample at a given water content. The desired water content of each soil sample was obtained by adding water to oven-dried soil. For Hokkaido loam, after repacking the soil samples at field water contents into 100 cm³ cores, the repacked soil cores were water saturated and then they were drained at different suctions using two different methods according to the matric suction ranges. A hanging water column was used for low matric suctions up to pF 2 (−100 cm H₂O) and a pressure plate apparatus for medium suctions (pF 2 to pF 4), where the pF equals $\log(-\psi)$ and ψ is the soil water matric potential in cm H₂O (− ψ cm H₂O is approximately equal to 0.1 ψ kPa). Finally, the repacked soil cores were air dried and the pF values of the samples were measured using a dew point potentiometer (WP4-T). The D_p was measured on the repacked 100 cm³ soil cores with a diffusion chamber method [Osozawa, 1987; Rolston and Moldrup, 2002]. Oxygen was used as tracer gas and measured as a function of time in the diffusion chamber. The D_p was calculated according to Osozawa [1987]. In this study, the gas diffusion coefficient of oxygen in free air ($D_{0,g}$) at 20°C equal to 0.20 (cm² s⁻¹) was used [Currie, 1960a; Glinski and Stepniewski, 1985].

4. Results and Discussion

4.1. Transport Parameters at Fluid Saturation

[21] Figure 2 shows the relative parameter values at fluid saturation (P_{sat}/P_0) for EC, D_p , and D_s as a function of total porosity (φ_{sat}) for artificial materials. The P_{sat}/P_0 for each parameter increased with increasing φ_{sat} . The P_{sat}/P_0 values for each parameter were well expressed by equation (1) with m ranging from 1.3 to 1.7. Especially, equation (1) with $m = 1.5$ as suggested by Marshall [1959] for D_p agreed well with the general trend of the measured EC, D_p , and D_s data.

[22] The P_{sat}/P_0 for EC, D_p , and D_s at fluid saturation for differently textured soils as a function of φ_{sat} and $\varphi_{sat} - \varphi_{th}^*$ are shown in Figures 3a and 3b, respectively. Similar to the artificial materials (Figure 2), the P_{sat}/P_0 values for each parameter for sandy soils were well expressed by equation (1) with m ranging from 1.3 to 1.7 (Figure 3a). The D_p data for water-filled aggregates from Jones et al. [2003], where intra-aggregate pores are filled with water and inter-aggregate pores are completely filled with air, were also well

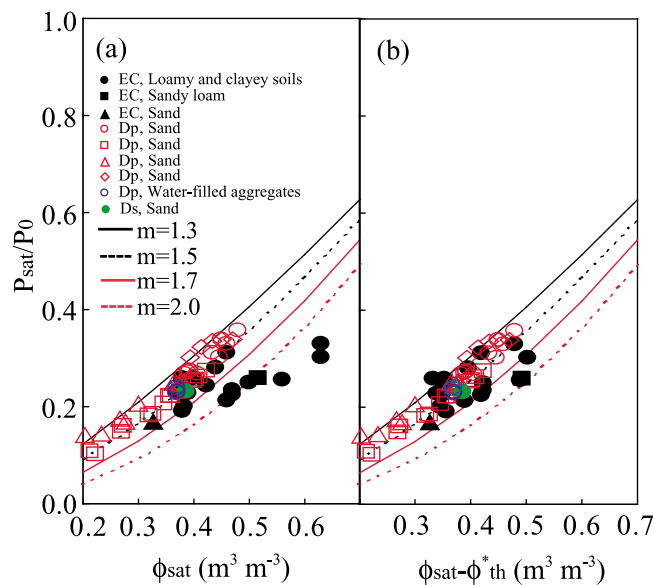


Figure 3. Relative parameter value, P/P_0 , for EC, D_p , and D_s for sandy, loamy, and clayey soils at fluid saturation as a function of (a) fluid saturation, ϕ_{sat} , and (b) active fluid content at saturation, $\phi_{\text{sat}} - \phi_{\text{th}}^*$. The solid and dotted lines represent the predictive curves by equation (1) in Figure 3a and by equation (3) in Figure 3b. Data are from *Mualem and Friedman* [1991], solid black circle; *Rhoades et al.* [1976], solid black square; *Weerts et al.* [1999], solid black triangle; *Hamamoto et al.* [2009a], open red circle; *Currie* [1960b], open red square; *Penman* [1940], open red triangle; *Taylor* [1949], open red diamond; *Jones et al.* [2003], open blue square; and *Hamamoto et al.* [2009b], solid green circle.

described by equation (1) with $m = 1.5$. This indicates that the water-filled aggregates exhibit a unimodal pore structure in the interaggregate pore space, likely similar to that for the unimodal sandy soils. The EC data for loamy and clayey soils showed lower P_{sat}/P_0 (i.e., higher m) at same ϕ_{sat} as compared with sandy soils for each parameter (Figure 3a). However, by considering a nonzero ϕ_{th}^* for EC data for loamy and clayey soils (0.02 – $0.20 \text{ m}^3 \text{ m}^{-3}$, ϕ_{th}^* expression of Table 1), all data for each parameter were well described by equation (3) with m ranging from 1.3 to 2.0 (Figure 3b). Overall, the excluded pore volume expansion of Archie's first law, equation (3), with m equal to 1.5 (ranging from 1.3 to 2.0) described well EC, D_p , and D_s data for variably textured soils at fluid saturation. Although our data suggest an m value of 1.5 to be in practice generally applicable for EC, D_p , and D_s data at fluid saturation, we note that *Hunt* [2005], *Hunt and Ewing* [2003], and *Ewing and Hunt* [2006] based on a percolation theory suggested an m value of around 2 for both EC and D_s . From percolation theory, *Hunt and Ewing* [2003] also derived a predictive D_p model analogous to the WLR (Marshall) model by *Moldrup et al.* [2000], finding an m value of around 1.5. A lack of experimental data prevents us at present from further investigating limitations or shortcomings of generally assuming $m = 1.5$ across transport parameters at fluid saturation.

4.2. Transport Parameters at Fluid-Unsaturated Conditions

[23] Figure 4 shows the normalized parameter values, P/P_{sat} (Table 1), for each parameter as a function of normalized ϕ values, $(\phi - \phi_{\text{th}})/(\phi_{\text{sat}} - \phi_{\text{th}})$, at fluid-unsaturated conditions. For this normalization, ϕ_{sat} was taken as the fluid content closest to soil total porosity and P_{sat} is the P value at ϕ_{sat} . Figure 4 documents the existence of percolation thresholds for each parameter, especially for D_s and TC, as compared with the case where the normalization assumed $\phi_{\text{th}} = 0$. In addition, the D_s and TC data showed significantly higher ϕ_{th} values for soils as compared with sands (Figures 4c–4d). This indicates that higher surface area for soils as compared with sands reduced water phase connectivity for D_s and water bridge effects on TC at dry conditions. By considering the ϕ_{th} values for each parameter defined in Table 1, P/P_{sat} for both sands and soils showed similar values at same normalized ϕ , indicating the proposed normalizations for both P and ϕ for each parameter are relevant across soil textures.

[24] Figure 5 shows the normalized P as a function of normalized ϕ for sands and soils, respectively. The EC, D_p , and D_s data showed similar P/P_{sat} at same normalized ϕ for both sands and soils. This analogy indicates that transport characteristics for EC, D_p , and D_s are equally affected by the pore structure (pore tortuosity and connectivity) in the effective fluid phase (air phase for D_p and water phase for EC and D_s) for both sands and soils. These findings seem highly promising for using any of the transport parameters to predict the other two parameters.

[25] For TC, the normalized TC data showed the expected “mirror image” effect (see Figure 1) as compared with the other three parameters. At low fluid content, the water bridge effects dramatically increase TC, while for EC, D_p , and D_s the transport is highly enhanced through well-connected, large-pore networks at high fluid contents. The modified Archie's second law considering percolation threshold (ϕ_{th}) (equation (4a) for EC, D_p , and D_s , or equation (5a) for TC) was tested against the P/P_{sat} for each parameter for sands and soils, respectively. Equation (4a) or (5a) with n as a fitting parameter could fit well measured data for each parameter (Figure 5). The best fit n values for each parameter and the model performance for $n = 1.75$, 2.0, and 2.5 for each parameter are summarized in Table 3. For all parameters, higher best fit n values for sands were obtained as compared with those for soils (Table 3). This is because at low ϕ condition, blocking effects on D_p by water-filled pores and on both EC and D_s by air-filled pores are higher (giving higher n) for sands containing larger-pore networks, while for TC the water bridge effects at low ϕ are higher (also giving higher n) for sands due to the lower soil surface area. The best fit n values for both TC and the three other parameters (EC, D_p , and D_s) were 2.0 and 2.2 for sands, and 1.8 and 2.1 for soils, respectively (Figure 5). However, for both sands and soils, equation (4a) or (5a) with $n = 2.0$ generally performed well against all parameters as shown in Table 3, which agrees well with the findings regarding the saturation exponent based on the percolation theory [*Hunt*, 2005]. Hence $m = 1.5$ (Figure 2) and $n = 2.0$ (Table 3) in equations (4a), (4b), (5a), or (5b) appear promising for universally predicting all four para-

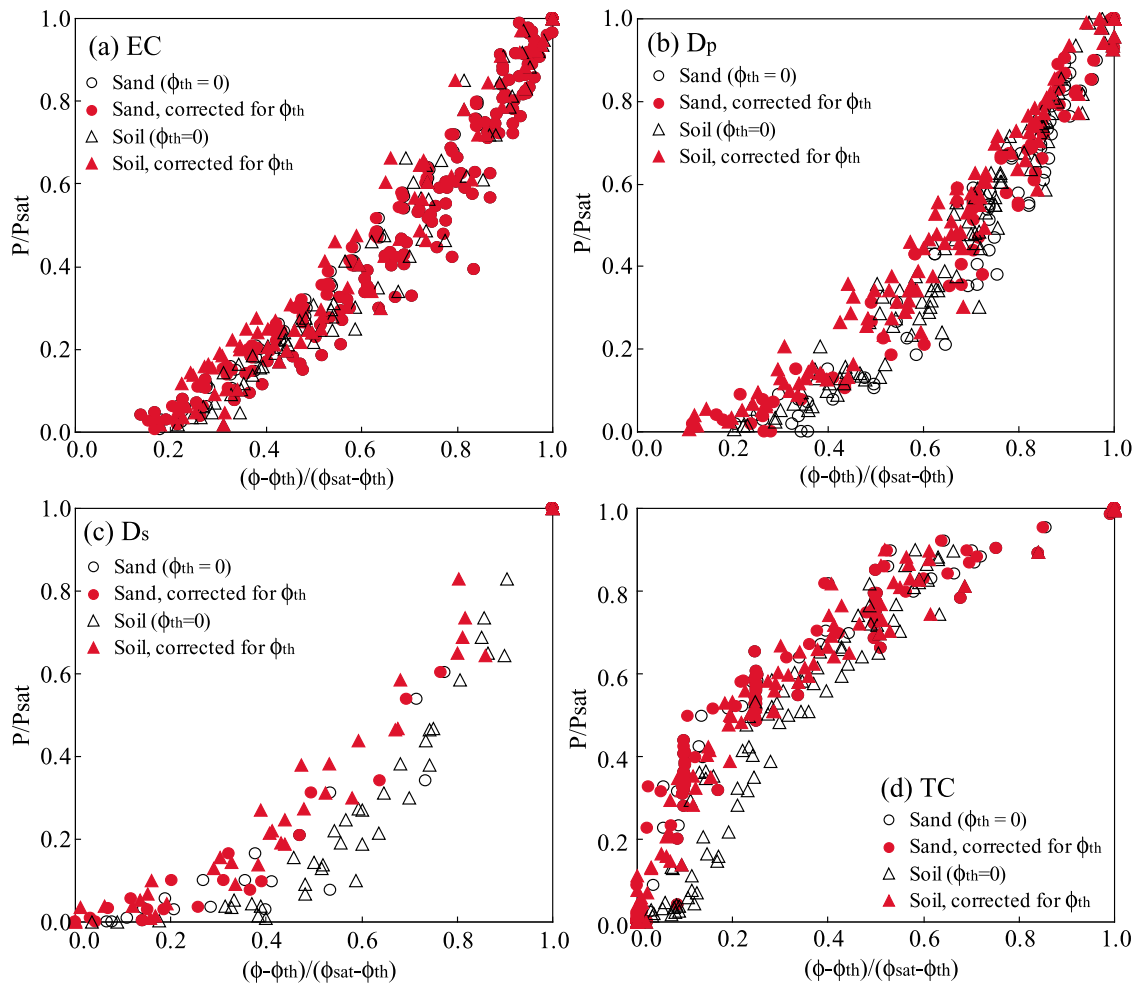


Figure 4. Normalized parameter value, P/P_{sat} , as a function of normalized φ , $(\varphi - \varphi_{\text{th}})/(\varphi_{\text{sat}} - \varphi_{\text{th}})$, for (a) electrical conductivity (EC), (b) D_p , (c) D_s , and (d) TC.

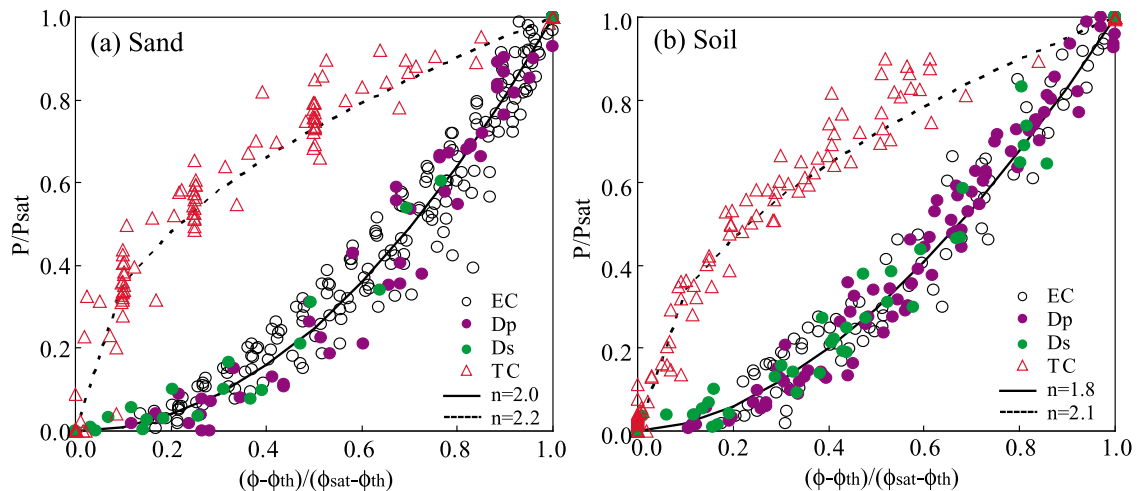


Figure 5. Normalized parameter value, P/P_{sat} , for EC, D_p , D_s , and TC as a function of normalized φ , $(\varphi - \varphi_{\text{th}})/(\varphi_{\text{sat}} - \varphi_{\text{th}})$, for (a) sands and (b) soils. The solid and dotted lines represent the fitted model curves against the combined data for EC, D_p , and D_s using equation (4a) and against TC data using equation (5a).

Table 3. Model Performance of the Modified Archie's Second Law, Equations (4a) and (5a)

		Optimal n		n = 1.75, RMSE ^a	n = 2.0, RMSE ^a	n = 2.5, RMSE ^a
		n	RMSE ^a			
EC	Sand	1.95	0.069	0.073	0.070	0.091
	Soil	1.68	0.063	0.064	0.079	0.121
	All	1.87	0.068	0.070	0.070	0.100
D _p	Sand	2.04	0.067	0.078	0.068	0.085
	Soil	1.76	0.056	0.056	0.066	0.111
	All	1.85	0.062	0.063	0.065	0.101
D _s	Sand	1.99	0.035	0.044	0.035	0.055
	Soil	1.72	0.046	0.046	0.058	0.096
	All	1.79	0.045	0.045	0.051	0.084
TC	Sand	2.28	0.059	0.090	0.068	0.063
	Soil	2.13	0.061	0.081	0.063	0.074
	All	2.21	0.061	0.086	0.066	0.068

^aRoot-mean-square error.

meters for variably textured soils at variably saturated conditions.

[26] Figure 6a shows normalized TC and D_p for sands at different bulk densities [Chen, 2008; Hamamoto et al., 2009a]. The TC and D_p data for each sand at different bulk densities were fitted by equations (4a) and (5a), respectively. The effects of soil compaction on the TC and D_p were not so clear (Figure 6a). However, higher n for both TC and D_p for highly compacted soils were observed as compared with loosely compacted soils. Soil compaction introduces higher possibility of creating a water bridge between closely aligned particles in sands, resulting in higher water bridge effects on TC (giving higher n) and higher water blockage effects on D_p (also giving higher n).

[27] Figure 6b shows the normalized TC and D_p for differently textured soils including organic soils. The dotted

and solid lines represent the fitted model curves by equations (4a) and (5a) against the TC and D_p data, respectively. The TC data for the sand showed higher n values than those for loamy and organic soils, indicating higher water bridge effects for the sand. This is because the easier formation of the water film with increasing water content due to the lower surface area enhanced the thermal conductivity through larger soil particles [Abu-Hamdeh and Reeder, 2000]. Conversely, the TC data for organic soils yielded the lowest n value (1.0). The thermal conductivity of organic matter ranges from 0.05 to 0.63 (W m⁻¹ K⁻¹) which is much lower than quartz (8.5 W m⁻¹ K⁻¹) and lower or similar to that of water (0.59 W m⁻¹ K⁻¹) [Sharratt, 1997; Jury and Horton, 2004]. Hence, for organic soils, the TC increased linearly corresponding to the incremental increase of the water content. The linear increase of TC with fluid content for organic soils has been also reported by Sharratt [1997]. For D_p data, no clear difference of n between the loamy soil and organic soil (mor) was observed. However, as already noted in Figure 5, a higher n value for the sand was observed due to the higher water blockage effects.

4.3. Independent Test of the Modified Archie's Second Law

[28] The modified Archie's second law (equations (4a) and (4b)) was tested against D_p and D_s data for Toyoura sand [Hamamoto et al., 2009a, 2009b], and Lerbjerg 3 clay loam and Lerbjerg 5 sandy clay [Moldrup et al., 2000; Olesen et al., 1999], whereas the bimodal extension of the modified Archie's second law (equations (6a) and (6b)) was tested against D_p data for Hokkaido loam with aggregate structure. The bulk density (ρ_b) and clay mass fraction (CF) for these soils are as follows: Toyoura sand (ρ_b = 1.58 Mg m⁻³,

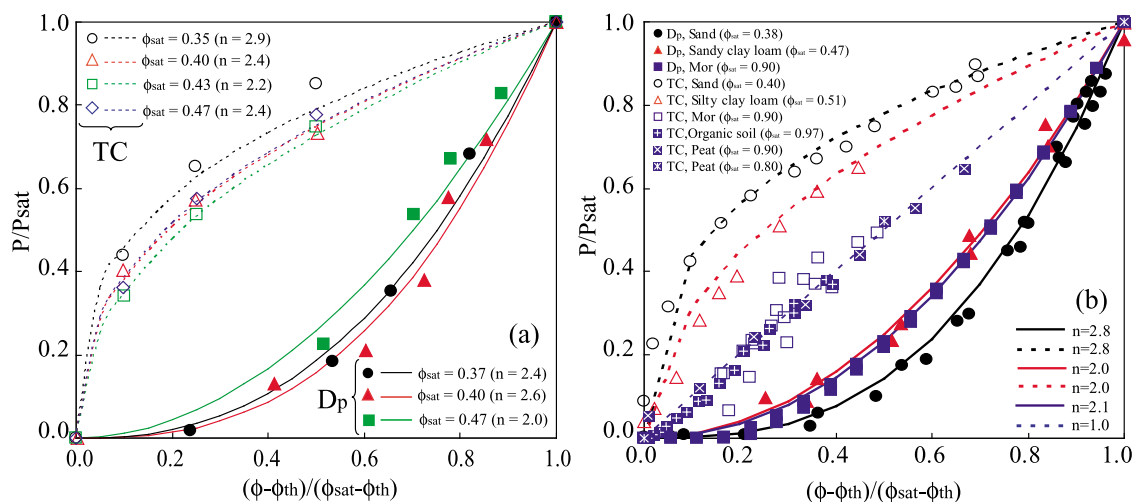


Figure 6. Normalized parameter value, P/P_{sat} , for D_p and TC as a function of normalized φ , $(\varphi - \varphi_{th})/(\varphi_{sat} - \varphi_{th})$, for (a) sands at different bulk densities and (b) differently textured soils. The solid and dotted lines represent the fitted model curves against the D_p using equation (4a) and against TC using equation (5a). The D_p and TC data in Figure 6a are from Hamamoto et al. [2009a] and Chen [2008]. Data in Figure 6b are from Currie [1961], solid black circle; Moldrup et al. [2000], solid red triangle; Lauren et al. [2000], solid blue square; Lu et al. [2007], open black square; Lu et al. [2007], open red triangle; Lauren et al. [2000], open blue square; Sharratt [1997], solid blue square with plus sign; Jury and Horton [2004], solid blue square with cross; and Bonan [2002], solid blue square with asterisk.

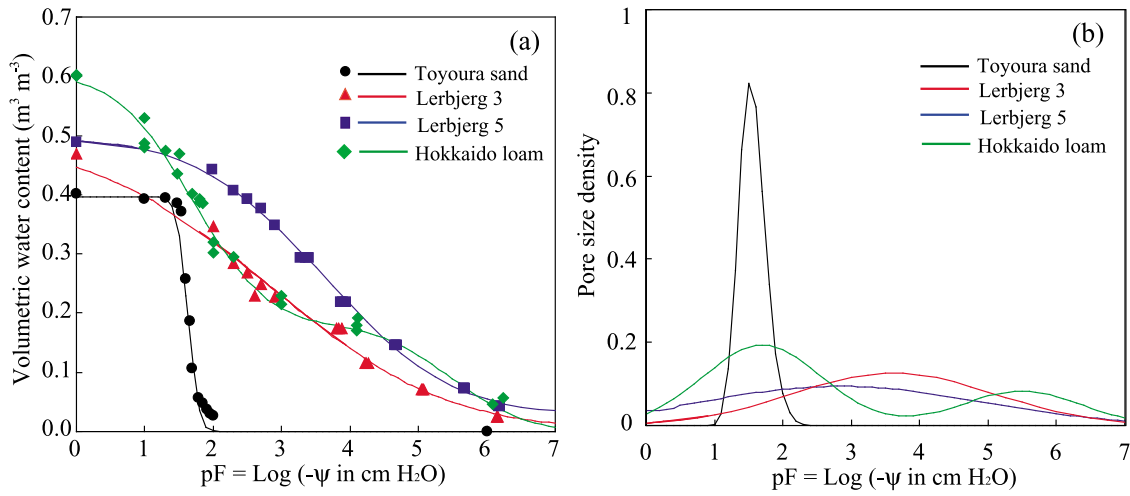


Figure 7. (a) Water retention curves and (b) pore size densities for Toyoura sand, Lerbjerg 3, Lerbjerg 5, and Hokkaido loam. The solid lines in Figure 7a represent fitted model curves using equation (8).

CF = 0.03), Lerbjerg 3 ($\rho_b = 1.46 \text{ Mg m}^{-3}$, CF = 0.21), Lerbjerg 5 ($\rho_b = 1.43 \text{ Mg m}^{-3}$, CF = 0.38), and Hokkaido loam ($\rho_b = 1.05 \text{ Mg m}^{-3}$, CF = 0.22) [Olesen *et al.*, 1999; Hamamoto *et al.*, 2009b].

[29] The soil water retention data for Toyoura sand and Hokkaido loam were obtained from Hamamoto *et al.* [2009b], and those for Lerbjerg 3 and Lerbjerg 5 were obtained from Olesen *et al.* [1999]. The water retention curves were fitted by a unimodal lognormal distribution model [Kosugi, 1996] for Toyoura sand, Lerbjerg 3, and Lerbjerg 5 and a bimodal lognormal distribution model [Seki, 2007] for Hokkaido loam, respectively. The bimodal lognormal distribution model [Seki, 2007] is

$$\phi_W = \phi_{W,r} + (\phi_{sat} - \phi_{W,r}) \sum_{i=1}^2 w_i Q \left[\frac{\ln(\psi/\psi_{mi})}{\sigma_i} \right], \quad w_1 + w_2 = 1, \quad (8)$$

where ψ_{mi} , σ_i , and w_i are fitting parameters, $Q(x)$ is the complementary cumulative normal distribution function, and $\phi_{W,r}$ is the residual water content ($\text{m}^3 \text{ m}^{-3}$). The unimodal lognormal distribution model corresponds to equation (8) with $i = 1$ [Kosugi, 1996]. The pore size density (PSD) is defined as the derivative of the water content with respect to $pF = \log(-\psi)$ [Durner, 1994; Seki, 2007],

$$PSD = [\ln(10)]\psi \sum_{i=1}^2 w_i \left(\frac{\phi_{sat} - \phi_{W,r}}{\sqrt{2\pi}\sigma_i\psi} \exp \left[-\frac{[\ln(\psi/\psi_{mi})]^2}{2\sigma_i^2} \right] \right). \quad (9)$$

The soil water retention curves for all soils are shown in Figure 7a. The unimodal lognormal distribution model (equation (8) with $i = 1$ [Kosugi, 1996]) and bimodal lognormal distribution (equation (8) with $i = 2$ [Seki, 2007]) represented well the water retention curves for each soil. Hokkaido loam showed two distinct regions of decrease in water content with increasing pF, indicating a bimodal pore structure due to aggregation. Figure 7b shows the pore size density for all soils. The pore size densities for the Toyoura sand, Lerbjerg 3, and Lerbjerg 5 showed one peak at around

pF 1.5, 3.0, and 3.8, respectively. For Hokkaido loam, the pore size densities showed two peaks at pF 1.5 and at pF 5.5, and the interaggregate and intra-aggregate pore regions were well separated at around pF 3.8.

[30] The D_p and D_s data for Toyoura sand, Lerbjerg 3, and Lerbjerg 5 were well described by the modified Archie's second law (equations (4a) and (4b)) as shown in Figure 8a. The fitted m and n values were again around 1.5 and 2.0, respectively. As previously suggested in Figure 5 and Table 3, the use of m equal to 1.5 and n equal to 2.0 seem applicable for predicting the general changes of transport parameters with fluid content for differently textured soils with a unimodal pore structure. Figure 8a showed markedly higher φ_{th} for D_s data for Lerbjerg 5 with higher clay content, while no obvious effects of soil texture on φ_{th} for D_p were observed. These differences in φ_{th} between D_p and D_s were well described by the predicted φ_{th} values based on the total porosity for D_p [Hunt, 2004b] (see Table 1) and the volumetric fines content for D_s [Moldrup *et al.*, 2007] (see Table 1).

[31] Figure 8b shows the measured and predicted relative D_p and D_s (P/P_0) for Hokkaido loam. Hokkaido loam exhibited a distinctly different behavior of D_p at the inter-aggregate ($\varphi < 0.4$) and intra-aggregate pore regions ($\varphi > 0.4$). The bimodal version of the modified Archie's second law (equations (6a) and (6b)) was tested against the D_p data for Hokkaido loam. For this, the interaggregate porosity (φ_1) was selected as 0.4, which is the soil air content corresponding to a pF of around 3.8 (Figure 7b), and a φ_{th} value equal to 0.04 was calculated from $0.1\varphi_1$ instead of $0.1\varphi_{sat}$ (Table 1). The m , n , and H values were fitted against the measured D_p data. The fitted m value (1.9) was higher than the average value previously obtained mainly from sand data ($m = 1.5$, Figure 2), agreeing well with the findings by Resurreccion *et al.* [2010] where tortuous inter-aggregate pore spaces (i.e., higher m) and $H = 0.5$ in the intra-aggregate region ($H = 0.6$ in this study) were reported for different types of aggregated soils.

[32] For D_s , the unimodal version of the modified Archie's second law (equations (4a) and (4b)) sufficed to describe measured data for Hokkaido loam. If it is assumed that the

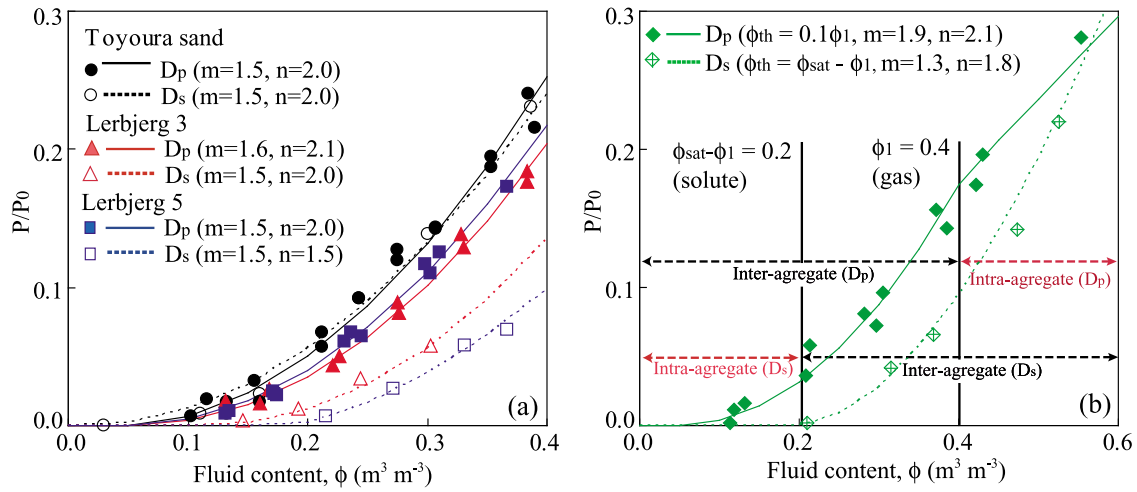


Figure 8. Relative parameter value, P/P_0 , for D_p and D_s for (a) Toyoura sand, Lerbjerg 3, and Lerbjerg 5, and (b) Hokkaido loam as a function of fluid content, ϕ . The solid and dotted lines in Figure 8a represent the fitted model curves against D_p and D_s using equation (4b). The solid line in Figure 8b represents the fitted model curve against D_p using equation (6b). The dotted lines in Figure 8b represent the fitted model curves against D_s using equation (4b) with ϕ_{th} taken as intra-aggregate pore volume, $\phi_{sat} - \phi_1$.

intra-aggregate pore space with smaller pore size is completely inactive for solute diffusion, only the interaggregate region need to be considered and the ϕ_{th} for aggregate soils is simply defined as the intra-aggregate pore volume. Figure 8b shows a good agreement between predicted D_s by equation (4b) with ϕ_{th} taken as the intra-aggregate pore volume ($\phi_{sat} - \phi_1$) and measured D_s . The fitted m was 1.3, which corresponds to the lower end of the expected range (Figure 2), and the fitted n was 1.8 close to the average value of 2.0 obtained for differently textured soils (Figure 5 and Table 3). Hence, by considering intra-aggregate pore space as ϕ_{th} , the modified Archie's second law seems applicable for predicting D_s in aggregated soils.

[33] Figure 9 shows the measured and predicted relative D_p and D_s (P/P_0) as a function of pF where the fluid content for each soil sample was converted to pF using water retention data (Figure 7a) and the fitted water relation function, equation (8). For D_p , Toyoura sand and Hokkaido loam showed a rapid increase of P/P_0 with increasing pF due to the drainage of soil water from the larger interparticle and interaggregate pores, respectively. Hokkaido loam showed two distinct regions of nonlinear increase of P/P_0 , corresponding to gas diffusion through interaggregate pore space and through both interaggregate and intra-aggregate pore space, respectively. Lerbjerg 3 and, more significantly, Lerbjerg 5 showed gradual increase of P/P_0 with pF because of the wider pore size distributions for both soils (Figure 7b). Conversely, for D_s , more rapid decreases of P/P_0 with increasing pF were observed for Toyoura sand and Hokkaido loam as compared with Lerbjerg 3 and Lerbjerg 5, due to the rapid reduction of water phase connectivity through larger-pore networks with increasing pF.

4.4. Some Distinct Features for Each Transport Parameter

[34] The analogy between the four parameters seems strong and promising, illustrated by the fact that the modi-

fied Archie's second law (equations (4a), (4b), (5a), or (5b)) generally describes/predicts well all four parameters for both sands and soils when applying $m = 1.5$ and $n = 2$. However, we note that all four parameters at the same time have distinct features and characteristics in their behavior as functions of fluid phase content. First, the percolation thresholds are more or less different for all four parameters,

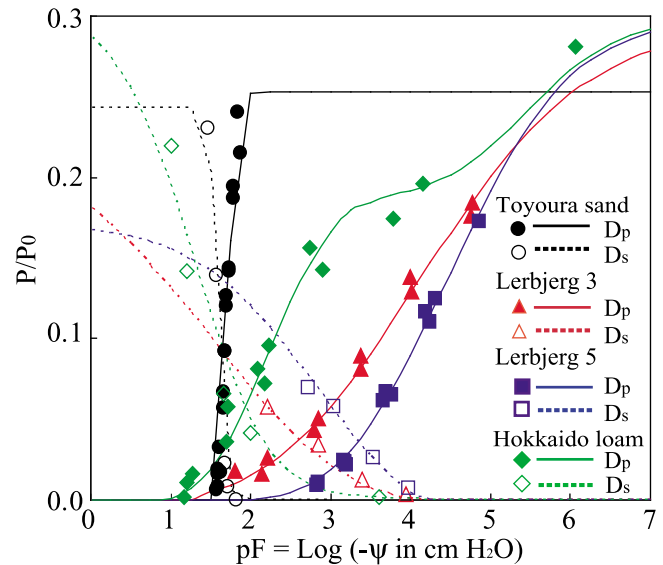


Figure 9. Relative parameter value, P/P_0 , for D_p and D_s as a function of pF. For Toyoura sand, Lerbjerg 3, and Lerbjerg 5, the solid and dotted lines represent the fitted model curves against D_p and D_s using equation (4b) in combination with equation (8) with $i = 1$. For Hokkaido loam, the solid and dotted lines represent the fitted model curves for D_p using equation (6b) and for D_s using equation (4b) in combination with equation (8) with $i = 2$, respectively.

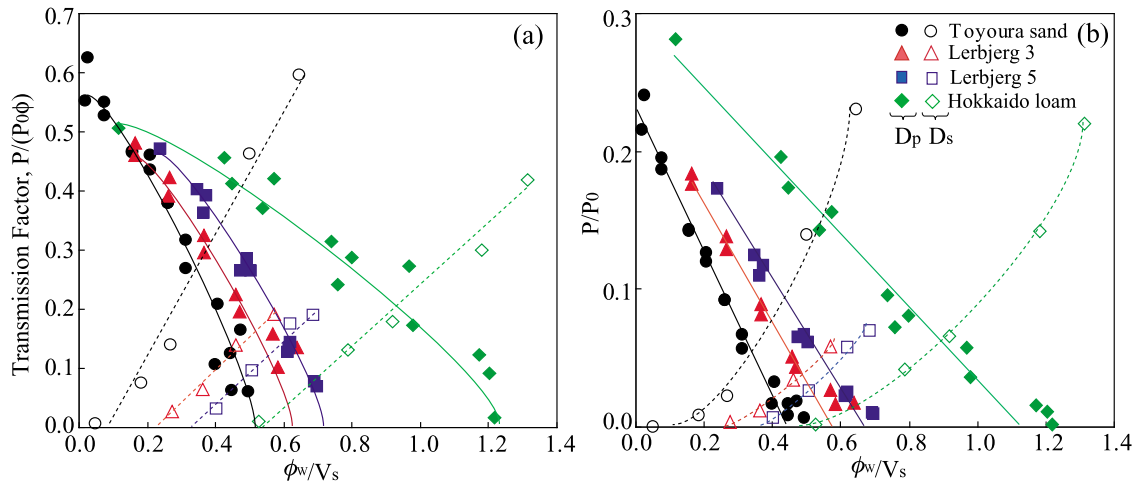


Figure 10. (a) Transmission factor, $P/(P_0\phi)$, and (b) relative parameter value, P/P_0 , for D_p and D_s as a function of ϕ_w/V_s for Toyoura sand, Lerbjerg 3, Lerbjerg 5, and Hokkaido loam.

as discussed and implied in Table 1. These formulations for the percolation thresholds need more attention for all four parameters in future studies to further improve applicability of the modified Archie's law. This will lead toward better understanding and prediction of each parameter for both separate and combined use, the latter, for example, in regard to soil thermal treatment where several of the parameters play a key role in optimal design and operation [Ho *et al.*, 1995; Saito *et al.*, 1998].

[35] Furthermore, the changes in the parameters with fluid phase content and the underlying fingerprints of saturated or partly saturated pore network tortuosity/connectivity for the individual phases (gas, liquid) will also exhibit distinct differences, as can be illustrated with D_p (gaseous phase) and D_s (liquid phase) as examples. Figure 10a shows the transmission factor ($P/P_0\phi$) as a function of ratio of volumetric water content (ϕ_w) to solid content (V_s). The ϕ_w/V_s and $P/P_0\phi$ represent the average water coverage surrounding soil particles and the gas or water phase connectivity, respectively. For D_p , $P/P_0\phi$ for each soil nonlinearly decreased with increasing ϕ_w/V_s . Especially Toyoura sand showed a more rapid decrease and lower $P/P_0\phi$ values were observed at same ϕ_w/V_s as compared with the finer-textured soils (Lerbjerg 3, Lerbjerg 5, Hokkaido loam). This indicates that the water menisci between larger-pore spaces (i.e., soil water in bottle necks) dramatically reduce gas diffusion with increasing water coverage. Conversely, $P/P_0\phi$ for D_s increased linearly with increasing ϕ_w/V_s and Toyoura sand exhibited a larger slope as compared with the other soils. The $P/P_0\phi$ for D_s showed higher values for Toyoura sand at the same ϕ_w/V_s due to a highly connected water phase through larger-pore networks. A linearity of the liquid phase transmission factor with ϕ_w has been also reported in previous studies [Porter *et al.*, 1960; Olesen *et al.*, 1999, 2000; Moldrup *et al.*, 2001, 2007]. As shown in Figure 8, the percolation threshold (ϕ_{th}) for D_s , which is highly related to volumetric contents of fines (Table 1), is a key factor to describe solute diffusion phenomena for differently textured soils.

[36] Figure 10b shows the relative parameter values (P/P_0) as a function of ϕ_w/V_s . The P/P_0 for D_s for each soil increased exponentially with increasing ϕ_w/V_s . However, the P/P_0 for D_p showed a linear decrease at drier condition and a nonlinear decrease at very wet condition for each soil. Mu *et al.* [2007] indicated that the relationship between the effective gas diffusivity and soil air content is strongly nonlinear when the soil air content (ϕ_{nw}) is less than a certain threshold value, and that relationship become quasi-linear for ϕ_{nw} more than the threshold value. The slope of P/P_0 with ϕ_w/V_s at drier condition for each soil, representing a connectivity of the air-filled pore networks in regard to gas diffusion, was highly related to the clay fraction (CF) of each soil (slope = $1.23 \text{ CF} - 0.56$, $R^2 = 0.96$). This strong dependency of soil texture on the gas phase connectivity for D_p , Figure 10b, and on the percolation threshold for D_s , Figure 8, clearly show a basic difference between gas and solute diffusion characteristics in soils. Thus, in addition to the suggested modified Archie's law model for predicting transport parameters based on a general analogy and difference between the four considered parameters, further investigations on each parameter are needed to further improve the model concept based on the specific characteristics for each transport phenomena.

5. Conclusions

[37] When introducing a percolation threshold (ϕ_{th}) for EC, D_p , and D_s , the normalized parameter values, P/P_{sat} , as a function of normalized fluid content, $(\phi - \phi_{th})/(\phi_{sat} - \phi_{th})$, suggested a strong analogy between these three parameters for both sandy and loamy-clayey soils. This analogy was well represented by an excluded volume expansion of Archie's second law, equations (4a) and (4b), including a cementation exponent (m) and a saturation exponent (n). In addition, TC was also well described by the modified Archie's second law if n^{-1} instead of n was applied, equations (5a) and (5b), in a "mirror image" approach representing the water-bridging effect on TC at low moisture contents.

[38] Higher n values were obtained for sands as compared with finer-textured soils, representing higher water blockage effects on D_p , air blockage effects on both EC and D_s , and higher water bridge effects on TC. However, $m = 1.5$ and $n = 2.0$ generally could describe well EC, D_p , D_s , and TC data for variably textured and variably saturated soils. This is useful for simulating multiphase transport phenomena including gas and solute diffusive and heat conductive transport in soils, and for predicting other parameter values from the easily measurable EC.

[39] Each parameter exhibits its own particular characteristics with changing fluid contents, illustrated by marked soil texture effects on the gas phase connectivity for D_p but on the percolation threshold for D_s , and general differences in percolation threshold for all four parameters. This calls for further studies where all four or more transport parameters are measured across moisture conditions on the same soils for further understanding the distinct transport characteristics of each parameter.

[40] **Acknowledgments.** This study was made possible by the Grant-in-Aid for Scientific Research 206192 and 18360224 from the Japan Society for the Promotion of Science (JSPS) and by a research grant from the Innovative Research Organization, Saitama University. This study was in part supported by the projects Gas Diffusivity in Intact Unsaturated Soil ("GADIUS") and Soil Infrastructure, Interfaces, and Translocation Processes in Inner Space ("Soil-It-Is") from the Danish Research Council for Technology and Production Sciences. We especially acknowledge the careful and dedicated laboratory work by a former B.S. student Akinori Toyokawa, Saitama University.

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